1) Explain the importance of statistics in business decision-making with suitable examples.

Here is a **more detailed and long explanation** of the importance of statistics in business decision-making with **elaborated examples** (useful for exams and assignments):

Importance of Statistics in Business Decision-Making (Detailed Explanation with Long Examples)

Statistics is a scientific method of **collecting, organizing, analyzing, and interpreting data**. In business, decisions must be taken carefully because wrong decisions can lead to losses. Statistics helps managers take **data-driven and logical decisions** instead of relying only on intuition.

Let us understand its importance in detail with long examples:

---

1. Planning and Forecasting

Statistics plays a vital role in **business planning** and **future forecasting**. By studying past data, businesses can predict future sales, demand, and growth trends.

Long Example:
A large automobile company wants to plan its production for the next year. It collects the sales data of the past 5 years and uses statistical tools like **trend analysis and time series analysis**. The data shows that the demand for electric vehicles has increased by 20% every year. Based on this statistical insight, the company decides to increase the production of electric cars and reduce the production of diesel vehicles. This helps the company avoid overproduction and stock wastage, and ensures customer demand is met.

---

2. Better Decision Making

Statistics helps managers evaluate different options and choose the best one based on facts and figures.

Long Example:
A retail chain wants to introduce a new product – organic food items. They conduct a market survey of 5,000 customers and collect data about income, buying habits, and preference for organic products. Statistical analysis shows that 65% of urban customers prefer organic food and are willing to pay a higher price. Based on these results, the company decides to introduce organic products only in urban stores instead of rural areas. Thus, statistics helps in making more accurate and profitable decisions.

---

3. Quality Control and Improvement

Statistics is used in manufacturing to maintain product quality by reducing defects and errors.

Long Example:
A smartphone manufacturing company uses statistical quality control techniques like **control charts and sampling methods**. Out of every 1,000 phones produced, a random sample of 100 phones is tested. Statistics shows that the defect rate is 3%, which is higher than the acceptable standard of 1%. The management studies the production process and identifies that a particular machine is causing screen defects. The machine is repaired or replaced, which reduces the defect rate in future production. This improves product quality and customer satisfaction.

---

4. Market Research and Customer Analysis

Statistics helps businesses understand customer behavior and market conditions.

Long Example:
A cosmetic company wants to launch a new fairness cream. Before launching, it conducts surveys in different age groups and income levels. After statistical analysis, it finds that women aged 18–25 prefer natural herbal products, while older customers prefer anti-aging products. Based on this data, the company designs different advertisements and packaging for each age group. Thus, statistics helps to target the right customers and increase sales.

---

5. Risk Management and Uncertainty Reduction

Business always involves risk. Statistics helps businesses measure and reduce risk.

Long Example:
A company wants to invest ₹10 crore in a new project. Before investing, the management studies statistical reports of similar projects, market demand, and economic trends. They calculate the probability of profit and loss using **probability theory**. The analysis shows a 75% chance of success and a 25% chance of failure. Based on this statistical risk assessment, the company modifies its investment plan to reduce possible losses.

---

6. Performance Evaluation

Statistics helps in measuring the performance of employees, departments, and business units.

Long Example:
A sales company collects data about monthly sales of each salesperson. Using statistical tools like averages and standard deviation, the manager finds that some employees consistently perform below average. The manager then arranges training programs for weak performers and reward programs for high performers. This improves overall productivity of the company.

---

7. Cost Control and Profit Maximization

Statistics helps identify unnecessary costs and increase efficiency.

Long Example:
A manufacturing company studies statistical data related to raw material wastage, machine downtime, and labor hours. The data shows that wastage is higher during night shifts. The company improves supervision and training during night shifts, which reduces wastage and increases profit.

Conclusion (Long)

In conclusion, statistics plays a **crucial role in business decision-making** because it converts raw data into meaningful information. It helps businesses in planning, forecasting, risk management, quality control, market research, and performance evaluation. Without statistics, business decisions would be based on guesswork, which can be risky. Therefore, statistics is an essential tool for achieving business growth, stability, and competitive advantage.

If you want, I can also provide this in **short points**, **MCQ format**, or **Marathi medium**.

2) . Define and differentiate between quantitative and qualitative data. Explain their types with suitable business-related examples.

Here is a **clear, detailed and exam-ready answer** to your question:

---

1. Definition of Quantitative and Qualitative Data

Quantitative Data

Quantitative data is **numerical data** that can be measured and counted. It is expressed in the form of numbers and can be analyzed using mathematical and statistical methods.

Examples in business:

- Monthly sales = ₹50,000
- Number of customers = 120
- Profit percentage = 12%

Qualitative Data

Qualitative data is **non-numerical data** that describes qualities, characteristics, or categories. It cannot be measured numerically and is usually expressed in words.

Examples in business:

- Customer satisfaction level (happy, neutral, unhappy)
- Brand image (good, average, poor)
- Type of product (electronics, clothing, groceries)

---

2. Differences Between Quantitative and Qualitative Data

Basis	Quantitative Data	Qualitative Data
Nature	Numerical	Descriptive (non-numerical)
Measurement	Can be measured and counted	Cannot be measured numerically
Analysis	Uses statistical and mathematical tools	Uses categorization and interpretation
Examples	Salary, sales, profit, age	Gender, color, brand type, customer opinion

---

3. Types of Quantitative Data (with Business Examples)

A. Discrete Data

Discrete data consists of **countable whole numbers**.

Business Example:

- Number of employees in a company = 150
- Number of products sold in a day = 85 units

(You cannot have 85.5 employees, so the value is fixed and countable.)

B. Continuous Data

Continuous data consists of **measurable values** that can take any value within a range, including decimals.

Business Example:

- Time taken to deliver a product = 2.5 hours
- Weight of raw material used = 1.75 kg
- Length of production time = 3.8 hours

---

4. Types of Qualitative Data (with Business Examples)

A. Nominal Data

Nominal data is categorized **without any order or ranking**.

Business Example:

- Type of products: Electronics, Furniture, Clothing
- Mode of payment: Cash, Credit Card, UPI

B. Ordinal Data

Ordinal data is categorized **with a meaningful order or ranking**, but the difference between categories is not measurable.

Business Example:

- Customer satisfaction: Very satisfied, Satisfied, Neutral, Dissatisfied
- Service quality rating: Excellent, Good, Average, Poor

---

5. Conclusion

Quantitative data is numerical and measurable, while qualitative data is descriptive and categorical. Both types of data are important in business because quantitative data helps in financial analysis and forecasting, while qualitative data helps in understanding customer behavior and preferences.

---

3) Statistics is the backbone of business decision-making." Discuss this statement in detail with real-world applications.

Here is a **detailed, long, and exam-ready answer** to your question:

"Statistics is the backbone of business decision-making." – Detailed Discussion with Real-World Applications

Statistics plays a **fundamental role in business decision-making** because it provides a scientific and systematic way to collect, analyze, interpret, and present data. In today's competitive business environment, decisions cannot be made based on guesswork or intuition alone. Statistics acts as the **backbone** because it supports every important business function such as planning, forecasting, marketing, finance, production, and human resource management.

Detailed discussion with real-world applications is given below:

---

1. Role of Statistics in Planning and Forecasting

Statistics helps businesses predict future trends by analyzing past and present data. This allows managers to prepare long-term and short-term plans.

Real-World Application:

Companies like **Amazon** and **Walmart** use historical sales data and statistical forecasting models to predict customer demand for products during festive seasons. For example, if data shows a sharp increase in online

shopping during Diwali, companies increase warehouse stock and delivery staff in advance. This avoids stock-outs and delayed deliveries.

---

2. Statistics in Market Research and Consumer Behavior
Businesses use statistics to understand customer needs, preferences, and buying behavior.
Real-World Application:
A company like **Coca-Cola** conducts surveys and collects data on taste preferences across different regions. Through statistical analysis, they found that some regions prefer less sugary drinks. Hence, Coca-Cola introduced products like **Coke Zero Sugar** and **Diet Coke** based on statistical insights.

---

3. Quality Control and Production Management
Statistics is used to maintain product quality and reduce manufacturing defects.
Real-World Application:
Automobile companies like **Toyota** use **Statistical Quality Control (SQC)** methods such as control charts. If defect rates exceed acceptable limits, production is stopped and machines are checked. This helps reduce recalls, warranty costs, and customer complaints.

---

4. Financial Decision-Making
Statistical tools help businesses in budgeting, investment decisions, and risk management.
Real-World Application:
Banks and financial institutions use **statistical models** to evaluate the creditworthiness of loan applicants. For instance, **HDFC Bank** or **SBI** analyze a customer's income history, repayment behavior, and credit score to statistically estimate the risk of loan default before approving a loan.

---

5. Human Resource Management
Statistics helps in employee performance evaluation, manpower planning, and training needs.
Real-World Application:
Large companies like **Infosys** track employee productivity through statistical performance metrics. If statistics show low productivity in a department,

management arranges skill development programs or restructures work processes.

---

6. Risk Analysis and Decision Under Uncertainty
Statistics helps to measure and manage business risks by calculating probabilities and outcomes.
Real-World Application:
Insurance companies like **LIC of India** use statistical probability and mortality tables to calculate premium amounts. These calculations ensure that the company can cover future claims and remain profitable.

---

7. Sales and Demand Analysis
Statistics helps companies understand sales trends and seasonal variations.
Real-World Application:
E-commerce platforms like **Flipkart** analyze customer search data, buying trends, and cart behavior statistically to decide which products to promote during big sales like **Big Billion Days**.

Conclusion (Exam-style)
Statistics is rightly called the **backbone of business decision-making** because it provides a solid foundation for rational and scientific decisions. It reduces uncertainty, improves accuracy, and increases business efficiency. Real-world companies across industries rely heavily on statistical methods for forecasting, quality control, financial planning, market research, and risk management. Without statistics, business decisions would be based on assumptions, increasing the chances of failure. Hence, statistics is essential for sustainable business growth and success.

---

4) The frequency distribution of monthly salaries (in ₹'000) of employees in a company is given below: Salary (₹'000) 10–20 20–30 30–40 40–50 50–60 Frequency 8 12 20 16 4 Calculate: (a) Mean salary (b) Median salary (c) Mode of the distribution

Here is the **step-by-step solution** to your question:

Given Data

Salary (₹'000)	Frequency (f)
10 – 20	8
20 – 30	12
30 – 40	20
40 – 50	16
50 – 60	4

Class width (h) = **10**

(a) Mean Salary

First find the **class midpoints (x)**:

Class	f	Midpoint (x)	f × x
10–20	8	15	120
20–30	12	25	300
30–40	20	35	700
40–50	16	45	720
50–60	4	55	220

Now calculate:

[
\sum f = 8+12+20+16+4 = 60
]

[
\sum fx = 120+300+700+720+220 = 2060
]

$$
\text{Mean} = \frac{\sum fx}{\sum f} = \frac{2060}{60} = 34.33
$$

✅ **Mean salary = ₹34.33 thousand**

(b) Median Salary

Formula:

$$
\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h
$$

Where:

- ($N = 60$)
- ($N/2 = 30$)

Find cumulative frequency (cf):

Class	f	Cumulative f
10–20	8	8
20–30	12	20
30–40	20	40 ✅ (Median class)
40–50	16	56
50–60	4	60

Median class = **30–40**

- ($l = 30$)
- ($cf = 20$)
- ($f = 20$)
- ($h = 10$)

Now substitute:

$$
\text{Median} = 30 + \left(\frac{30 - 20}{20}\right)\times10
$$
$$
= 30 + \left(\frac{10}{20}\right)\times10
$$
$$
= 30 + 5 = 35
$$

✅ **Median salary = ₹35 thousand**

(c) Mode of the Distribution

Formula:

$$
\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right)\times h
$$

Modal class = class with highest frequency = **30–40**

Values:

- $(l = 30)$
- $(f_1 = 20)$
- $(f_0 = 12)$ (previous class)
- $(f_2 = 16)$ (next class)
- $(h = 10)$

Now calculate:

$$
\text{Mode} = 30 + \left(\frac{20 - 12}{2(20) - 12 -16}\right)\times10
$$
$$
= 30 + \left(\frac{8}{40 -28}\right)\times10
$$

$$
= 30 + \left(\frac{8}{12}\right)\times10
$$

$$
= 30 + 6.67 = 36.67
$$

✅ **Mode salary = ₹36.67 thousand**

✅ **Final Answers**

Measure	Answer
Mean	**₹34.33 thousand**
Median	**₹35 thousand**
Mode	**₹36.67 thousand**

If you want, I can also show you **shortcut methods** for these calculations.

5) The following data represent the profit (in ₹ lakh) of 8 firms: 12, 16, 14, 18, 22, 24, 20, 26 Calculate: (a) Mean profit (b) Mean deviation from the mean (c) Coefficient of mean deviation

Let's do it step by step 👇

Given profits (₹ lakh):
12, 16, 14, 18, 22, 24, 20, 26
Number of firms, (n = 8)

(a) Mean Profit

$$
\bar{x} = \frac{\sum x_i}{n}
$$

$$
\sum x_i = 12 + 16 + 14 + 18 + 22 + 24 + 20 + 26
$$

Add them:

- $(12 + 16 = 28)$
- $(28 + 14 = 42)$
- $(42 + 18 = 60)$
- $(60 + 22 = 82)$
- $(82 + 24 = 106)$
- $(106 + 20 = 126)$
- $(126 + 26 = 152)$

So,

$$
\bar{x} = \frac{152}{8} = 19
$$

✅ **Mean profit = ₹19 lakh**

(b) Mean Deviation from the Mean

Mean deviation from mean:

$$
\text{MD} = \frac{\sum |x_i - \bar{x}|}{n}
$$

First, calculate deviations from mean (19):

Profit (x_i)	(x_i - 19)	($|x_i - 19|$)
12	$12 - 19 = -7$	7
16	$16 - 19 = -3$	3
14	$14 - 19 = -5$	5
18	$18 - 19 = -1$	1

22	22 − 19 = 3	3
24	24 − 19 = 5	5
20	20 − 19 = 1	1
26	26 − 19 = 7	7

Now sum of absolute deviations:

$$
7 + 3 + 5 + 1 + 3 + 5 + 1 + 7 = 32
$$

So,

$$
\text{MD} = \frac{32}{8} = 4
$$

✅ **Mean deviation from mean = 4 lakh**

(c) Coefficient of Mean Deviation

Coefficient of mean deviation (based on mean):

$$
\text{Coefficient of MD} = \frac{\text{Mean Deviation}}{\text{Mean}} = \frac{4}{19} \approx 0.21
$$

✅ **Coefficient of mean deviation ≈ 0.21**

Final Answers

- (a) **Mean profit = ₹19 lakh**
- (b) **Mean deviation from mean = ₹4 lakh**
- (c) **Coefficient of mean deviation ≈ 0.21** (or 21%)

6) *5. A firm recorded the production output (units) for 6 days as: 120, 135, 140, 150, 160, 155 Calculate mean, range, variance, and standard deviation.

Let’s solve it step by step in a clear and exam-ready way.

Given Data (units produced in 6 days)

120, 135, 140, 150, 160, 155

Number of observations, $n = 6$

---

1. Mean (Average)

$$\text{Mean} = \frac{\sum x}{n}$$

$$\sum x = 120 + 135 + 140 + 150 + 160 + 155$$

Add them:

120 + 135 = 255
255 + 140 = 395
395 + 150 = 545
545 + 160 = 705
705 + 155 = 860

$$\text{Mean} = \frac{860}{6} = 143.33$$

✅ **Mean = 143.33 units**

---

2. Range

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Max = 160
Min = 120

$$
\text{Range} = 160 - 120 = 40
$$

✅ **Range = 40 units**

---

3. Variance

Formula:

$$
\sigma^{2} = \frac{\sum (x - \bar{x})^{2}}{n}
$$

First, find deviations from the mean (143.33):

(x)	(x-\bar{x})	((x-\bar{x})^2)
120	-23.33	544.29
135	-8.33	69.39
140	-3.33	11.09
150	6.67	44.49
160	16.67	277.89
155	11.67	136.19

Now sum of squared deviations:

$$
544.29 + 69.39 + 11.09 + 44.49 + 277.89 + 136.19 = 1083.34
$$

$$
\text{Variance} = \frac{1083.34}{6} = 180.56
$$

✅ **Variance ≈ 180.56**

---

4. Standard Deviation

$$
\sigma = \sqrt{\text{Variance}} = \sqrt{180.56}
$$

$$
\sigma \approx 13.44
$$

✅ **Standard Deviation ≈ 13.44 units**

✅ **Final Answers**

Measure	Result
Mean	**143.33 units**
Range	**40 units**
Variance	**180.56**
Standard Deviation	**13.44 units**

If you want, I can show you a **shortcut method** for variance and standard deviation too.

7) What is the Binomial distribution? State its assumptions, characteristics, and derive its mean and variance. Provide real business applications.

Here is a **detailed, long and exam-ready answer** to your question on the **Binomial Distribution**, with derivation of **mean, variance**, assumptions, characteristics and **real business applications**:

Binomial Distribution – Detailed Explanation

1. What is Binomial Distribution?

A **Binomial distribution** is a discrete probability distribution that represents the number of **successes** in a fixed number of **independent trials**, where:

- each trial has only two possible outcomes: **success** or **failure**
- the probability of success remains **constant** in all trials

In simple words, it answers questions like:

"Out of *n* trials, how many times does a success occur?"

Example (simple):
If a company sends emails to 10 customers and the probability that a customer opens an email is 0.4, the binomial distribution helps find the probability that exactly 5 customers open the email.

---

2. Assumptions of Binomial Distribution

For a situation to follow a binomial distribution, the following **assumptions must be satisfied**:

1. **Fixed number of trials (n)**
   The number of trials must be predetermined and finite.
2. **Only two outcomes per trial**
   Each trial results in either:
   - Success (S)
   - Failure (F)
3. **Constant probability of success (p)**
   The probability of success remains the same in every trial.
4. **Independence of trials**
   The result of one trial does not affect the result of another.
5. **Random process**
   Trials are conducted randomly.

---

3. Characteristics (Properties) of Binomial Distribution

1. It is a **discrete probability distribution**.
2. It depends on two parameters:
   - (n) = number of trials
   - (p) = probability of success
     (with (q = 1 - p) = probability of failure)
3. The graph of the binomial distribution can be:
   - Symmetrical when (p = 0.5)
   - Positively skewed when (p < 0.5)
   - Negatively skewed when (p > 0.5)
4. Mean and variance depend on both (n) and (p).
5. It is used when outcomes are based on **Yes/No type** events.

---

4. Binomial Distribution Formula

The probability of getting exactly (x) successes in (n) trials is:

[
P(X = x) = \binom{n}{x} p^x q^{,n-x}
]

Where:

- (\binom{n}{x} = \frac{n!}{x!(n-x)!})
- (p) = probability of success
- (q = 1 - p)
- (x) = number of successes

---

5. Derivation of Mean of Binomial Distribution

Let (X) be the number of successes in (n) independent trials.

We consider:

- $X = X_1 + X_2 + X_3 + \dots + X_n$
  where each X_i is a Bernoulli random variable.

For each trial:

$$E(X_i) = p$$

Then, by **linearity of expectation**:

$$E(X) = E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$E(X) = p + p + \dots + p = np$$

✅ **Mean of Binomial Distribution:**

$$\mu = np$$

---

6. Derivation of Variance of Binomial Distribution

Variance of each Bernoulli trial:

$$Var(X_i) = p(1-p) = pq$$

Total variance:

$$Var(X) = Var(X_1 + X_2 + \dots + X_n)$$

Since trials are independent:

$$
Var(X) = Var(X_1) + Var(X_2) + ... + Var(X_n)
$$

$$
Var(X) = pq + pq + ... + pq = npq
$$

✅ **Variance of Binomial Distribution:**

$$
\sigma^2 = npq
$$

✅ **Standard Deviation:**

$$
\sigma = \sqrt{npq}
$$

---

7. Real Business Applications of Binomial Distribution (Long Examples)

Example 1: Quality Control in Manufacturing

A mobile phone manufacturing company wants to estimate how many phones will be defective.

- Total phones produced in a batch = 100
- Probability that a phone is defective = 0.05

Using the binomial distribution, the company can calculate the probability of getting:

- exactly 2 defective phones
- more than 5 defective phones

This helps the company decide:

- whether to accept or reject the batch
- whether to improve the production process

Example 2: Marketing Campaign Success

An e-commerce company sends promotional emails to 50,000 customers.

Past data shows:

- Probability of a customer clicking the link = 0.10

The company uses the binomial distribution to estimate:

- How many customers are likely to click the link
- The probability that at least 6,000 customers will respond

This helps in planning inventory, discounts, and server capacity.

Example 3: Loan Approval in Banking

A bank studies past loan data and finds:

- Probability that a customer defaults on a loan = 0.08

If the bank gives loans to 200 customers, it uses the binomial model to estimate:

- expected number of defaulters
- risk level of its loan portfolio

This helps the bank in **risk management and policy making**.

Example 4: Hiring Decisions in HR

A company conducts online aptitude tests for 500 candidates.

Previous statistics show:

- Probability that a candidate clears the test = 0.25

The HR department uses binomial distribution to estimate:

- expected number of candidates who will pass
- probability that more than 150 candidates will qualify

This helps them plan interview slots and onboarding.

Example 5: Inventory and Demand Prediction

A retail store knows:

- Probability that a customer buys a particular product = 0.3

For 100 walk-in customers, binomial distribution helps estimate:

- how many sales will happen
- probability of stock-out

This ensures proper **inventory management**.

---

8. Conclusion (Exam Style)

The binomial distribution is a powerful statistical tool that models real-life business situations where outcomes are binary in nature such as success/failure, defective/non-defective, click/no click, and pass/fail. Its assumptions ensure reliability, while its derived mean (np) and variance (npq) provide useful measures of expected performance and variability. Due to its wide applicability in quality control, marketing, finance, HR, and operations, binomial distribution plays a vital role in business decision-making.

If you want, I can also give:

- Short note version
- Formula sheet
- Numerical problems with solutions
- Marathi explanation

8) Describe the Normal distribution. Explain its properties and how it is applied in business forecasting and quality control.

Below is a **detailed, long, and exam-ready answer** for your question on the **Normal Distribution**, its properties, and applications in business.

Normal Distribution – Explanation, Properties, and Business Applications

1. What is Normal Distribution?

The **Normal distribution**, also called the **Gaussian distribution** or **bell-shaped curve**, is one of the most important probability distributions in statistics. It describes how data is distributed around a central value (mean).

It is **continuous**, **symmetrical**, and shaped like a bell.

Real Meaning

Many natural and business-related variables follow the Normal distribution, such as:

- heights of people
- IQ scores
- monthly sales figures
- product measurements
- employees' performance rating

Probability Density Function (PDF)

The general equation of a normal distribution is:

$$
f(x)=\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}
$$

Where:

- μ = mean
- σ = standard deviation
- σ^2 = variance

---

2. Properties (Characteristics) of Normal Distribution

Below are the key properties (write these in exams):

1. Bell-Shaped Curve

The curve is smooth, symmetric, and bell-shaped.

2. Symmetrical Around the Mean

Mean = Median = Mode
All located at the center of the distribution.

3. Total Area = 1

The entire area under the curve is 1 or 100%.

4. Tails Extend Infinitely

Both ends approach zero but never touch the X-axis.

5. Defined by Two Parameters Only

A normal distribution depends only on:

- **Mean (μ)** → location of center
- **Standard deviation (σ)** → spread/width of distribution

6. Empirical Rule (68–95–99.7 Rule)

This is the most important property:

- **68%** of data lies within **1σ** of mean
- **95%** of data lies within **2σ** of mean
- **99.7%** of data lies within **3σ** of mean

This rule is used heavily in business forecasting and quality control.

---

3. Why is the Normal Distribution Important?

- Many real-world phenomena naturally follow the normal distribution.
- Calculations of probability become easy.

- Forms the basis for many statistical techniques (e.g., confidence intervals, hypothesis testing).

---

4. Applications in Business

A. Normal Distribution in Business Forecasting

Business forecasting involves predicting future values like sales, demand, revenue, and production. Normal distribution helps in making these predictions reliable.

How It Helps:

1. Demand Forecasting

Businesses can estimate future demand for products.

Example (Detailed):
A grocery retailer observes that daily demand for milk is normally distributed with:

- Mean = 500 liters
- Standard deviation = 50 liters

Using the normal distribution, the manager can estimate:

- probability that demand will exceed 600 liters
- how much stock to keep to avoid stock-outs
- weekly and monthly forecasting

This prevents overstocking or understocking.

---

2. Sales Forecasting

Companies use past normally distributed sales data to predict:

- future monthly sales
- seasonal variations
- required manpower and resources

Example:
A clothing company uses historical sales data of shirts and finds it follows a normal pattern. They predict holiday-season sales and plan production accordingly.

---

3. Financial Forecasting

Share price fluctuations, return on investments, and interest rates often follow an approximately normal distribution.

Banks and financial analysts use it for:

- risk assessment
- profit prediction
- value at risk (VaR)

B. Normal Distribution in Quality Control

Normal distribution is widely used in manufacturing industries to maintain and improve quality.

1. Control Charts (QC Tools)

Companies use control charts to check whether production is stable or not.

If measurements (length, weight, diameter, tolerance) are normally distributed:

- 99.7% of products fall within $\pm 3\sigma$
- Any product outside $\pm 3\sigma$ indicates a defect

Example (Detailed):
An automobile company manufactures pistons with diameter:

- Mean = 5.00 cm
- $\sigma = 0.02$ cm

Using normal distribution:

- Acceptable range = 5 ± 0.06 (3σ rule)
- Any piston outside this range is rejected

This ensures **consistency**, **lower defect rates**, and **high customer satisfaction**.

---

2. Six Sigma Quality Method

Six Sigma is based entirely on the normal distribution.

- A Six Sigma process tries to reduce defects to **3.4 per million products**.
- This is achieved by reducing the spread (σ) of the distribution around the target value.

---

3. Setting Specifications and Tolerances

Manufacturers set acceptable upper and lower limits using the normal curve.

Example:
A pen factory produces refills of length 14 cm with $\sigma = 0.1$ cm.
They know 99.7% products fall within 14 ± 0.3 cm.
So they set tolerance limits:
13.7 cm to 14.3 cm

This reduces customer complaints.

---

4. Acceptance Sampling

Using normal distribution, a company can:

- test a small sample
- check if the whole batch is acceptable

Instead of testing each product.

---

5. Conclusion (Exam Style)

The normal distribution is one of the most important statistical distributions because of its simplicity, symmetry, and universal applicability. It describes how data naturally clusters around the mean. Its properties such as 68–95–99.7 rule make it a powerful tool in business. In forecasting, it helps predict future sales,

demand, and financial performance. In quality control, it ensures that products meet specifications and helps identify defects through control charts, Six Sigma, and tolerance setting. Therefore, normal distribution forms the backbone of decision-making in modern business operations.

---

9) A manufacturer knows that 10% of its products are defective. A sample of 12 items is chosen at random. Calculate: (a) The probability that exactly 3 items are defective (b) The probability that at most 2 items are defective (c) The mean and variance of the distribution

We'll treat this as a **binomial distribution** problem.

Let

- (X) = number of defective items in a sample of 12
- (n = 12) (trials)
- (p = 0.10) (probability an item is defective)
- (q = 1 - p = 0.90)

Binomial probability formula:

[
P(X = x) = \binom{n}{x} p^x q^{n-x}
]

(a) Probability that exactly 3 items are defective

[
P(X = 3) = \binom{12}{3} (0.10)^3 (0.90)^9
]

[
\binom{12}{3} = \frac{12!}{3! , 9!} = 220
]

$$
P(X = 3) = 220 \times (0.10)^3 \times (0.90)^9
$$

$$
= 220 \times 0.001 \times 0.3874 \approx 0.0852
$$

✅ **$P(X = 3) \approx 0.0852$**

(b) Probability that at most 2 items are defective

"At most 2" means: $(X = 0, 1, 2)$

$$
P(X \le 2) = P(0) + P(1) + P(2)
$$

$(P(0))$:

$$
P(0) = \binom{12}{0}(0.10)^0(0.90)^{12} = 1 \cdot 1 \cdot 0.90^{12} \approx 0.2824
$$

$(P(1))$:

$$
P(1) = \binom{12}{1}(0.10)^1(0.90)^{11} = 12 \cdot 0.10 \cdot 0.90^{11} \approx 0.3766
$$

$(P(2))$:

$$
P(2) = \binom{12}{2}(0.10)^2(0.90)^{10} = 66 \cdot 0.01 \cdot 0.90^{10} \approx 0.2301
$$

Now add:

[
P(X \le 2) \approx 0.2824 + 0.3766 + 0.2301 \approx 0.8891
]

✅ **(P(\text{at most 2 defective}) \approx 0.8891)**

(c) Mean and Variance of the Distribution

For a binomial distribution:

- **Mean** (\mu = np)
- **Variance** (\sigma^2 = npq)

Mean:

[
\mu = np = 12 \times 0.10 = 1.2
]

Variance:

[
\sigma^2 = npq = 12 \times 0.10 \times 0.90 = 1.08
]

(If needed, **standard deviation** (\sigma = \sqrt{1.08} \approx 1.04))

Final Answers

- (a) (P(\text{exactly 3 defective}) \approx \mathbf{0.0852})
- (b) (P(\text{at most 2 defective}) \approx \mathbf{0.8891})
- (c) Mean (= \mathbf{1.2}), Variance (= \mathbf{1.08})

10) For events A and B $P(A) = 0.45$, $P(B) = 0.55$, $P(A \cup B) = 0.80$ Find:
(a) $P(A \cap B)$ (b) $P(A\ c \cap B)$ (c) $P(A \mid B)$

Given:

[
P(A)=0.45,\quad P(B)=0.55,\quad P(A\cup B)=0.80
]

(a) Find (P(A \cap B))

Use the formula:
[
P(A\cup B)=P(A)+P(B)-P(A\cap B)
]

[
0.80=0.45+0.55-P(A\cap B)
]
[
P(A\cap B)=1.00-0.80=0.20
]

✅ **Answer:** (P(A \cap B)=0.20)

(b) Find (P(A^c \cap B))

This means "B happens but A does not".

[
P(A^c \cap B)=P(B)-P(A\cap B)
]
[
=0.55-0.20=0.35
]

✅ **Answer:** (P(A^c \cap B)=0.35)

(c) Find (P(A \mid B))

Formula for conditional probability:
[
P(A\mid B)=\frac{P(A\cap B)}{P(B)}

]
[
=\frac{0.20}{0.55}=\frac{20}{55}=\frac{4}{11}\approx0.3636
]

✅ **Answer:**
[
P(A\mid B)=\frac{4}{11}\approx0.364
]

Final Answers Summary:

(a) (P(A\cap B)=0.20)
(b) (P(A^c\cap B)=0.35)
(c) (P(A\mid B)=0.364)

If you want a simple diagram explanation, just tell me! 😊

11) A company sells a daily-use product whose demand is normally distributed with mean = 600 units and SD = 100 units. Management wants to stock enough units so that demand is met 95% of the time. Find: (a) The stock level required (b) The required safety stock (Use Z = 1.645)

Given:

- Mean demand (\mu = 600) units
- Standard deviation (\sigma = 100) units
- Service level = **95%**
- Given **Z = 1.645**

(a) Stock level required

Formula:

$$
\text{Stock level} = \mu + Z\sigma
$$

$$
= 600 + (1.645)(100)
$$

$$
= 600 + 164.5 = 764.5
$$

✅ **Stock level required ≈ 765 units**

(b) Required Safety Stock

Formula:

$$
\text{Safety Stock} = Z\sigma
$$

$$
= 1.645 \times 100 = 164.5
$$

✅ **Safety stock ≈ 165 units**

Final Answers:

Part	Answer
(a) Stock level required	≈ **765 units**
(b) Safety stock	≈ **165 units**

If you want a quick explanation in simple words, just ask 😊

12) Explain the steps involved in Hypothesis Testing. Discuss concepts such as null hypothesis, alternative hypothesis, Type I error, Type II error, and level of significance with examples.

Here's a **clear and simple explanation** of Hypothesis Testing and its main concepts:

✅ What is Hypothesis Testing?

Hypothesis testing is a statistical method used to make decisions about a population based on sample data.

In simple words:

It helps us decide whether a claim is **true or false** using data.

✅ Steps Involved in Hypothesis Testing

Here are the main steps:

1. State the Hypotheses

You form two statements:

- **Null Hypothesis (H_0):**
  The statement that there is **no change, no effect, or no difference**.
- **Alternative Hypothesis (H_1 or H_a):**
  The statement that **contradicts the null hypothesis**.

Example:
Suppose a company claims that the average weight of a soap bar is 100g.

- H_0: $\mu = 100$ (The soap weight is 100g)
- H_1: $\mu \neq 100$ (The soap weight is not 100g)

---

2. Choose the Level of Significance (α)

The **level of significance** is the probability of making a **Type I error**.

Common values:

- $\alpha = 0.05$ (5%)
- $\alpha = 0.01$ (1%)

Example:
Choosing $\alpha = 0.05$ means:

You allow a 5% chance of wrongly rejecting a true null hypothesis.

---

3. Collect Sample Data

You take a sample from the population and compute:

- Sample mean
- Standard deviation
- Test statistic (like Z-test or t-test value)

---

4. Calculate the Test Statistic

Use the appropriate formula depending on the test.

Example for Z-test:

$$
Z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}
$$

Where:

- $\bar{x}$ = Sample mean
- μ = Population mean
- σ = Population standard deviation
- n = Sample size

---

5. Make a Decision

Compare:

- Calculated test value **vs**
- Critical value (from Z or t table)

Two outcomes:
✅ Reject (H_0)
❌ Fail to reject (H_0)

---

6. Draw a Conclusion

State the result in plain language.

Example:

There is enough evidence to conclude that the average soap weight is different from 100g.

✅ Important Concepts

1. Null Hypothesis (H_0)

- A default statement
- Assumes **no effect or no difference**

Example:
H_0: The new medicine has no effect.

---

2. Alternative Hypothesis (H_1)

- Opposite of H_0
- Says there **is an effect or difference**

Example:
H_1: The new medicine is effective.

---

3. Type I Error

- Rejecting a **true null hypothesis**
- Called a **false positive**

Example:
Saying “the medicine works” when actually it doesn’t.

Probability:
[
P(\text{Type I error})=\alpha
]

---

4. Type II Error

- Accepting (failing to reject) a **false null hypothesis**
- Called a **false negative**

Example:
Saying “the medicine doesn’t work” when it actually does.

Probability:
[
P(\text{Type II error})=\beta
]

---

5. Level of Significance (α)

- The maximum risk of making a **Type I error** that you are willing to take.
- Common values: 0.05, 0.01

Example:
If α = 0.05 → You accept a 5% risk of wrong rejection.

✅ **Simple Example (Real Life)**

Scenario:

A school claims that the average exam marks are 70.

- **H_0:** $\mu = 70$
- **H_1:** $\mu \neq 70$
- Choose $\alpha = 0.05$
- Sample mean found = 75

If the test result falls in the rejection region → Reject H_0.

✅ Summary Table

Concept	Meaning
Null Hypothesis (H_0)	Statement of no effect
Alternative Hypothesis (H_1)	Statement of effect
Type I Error	Rejecting a true H_0
Type II Error	Accepting a false H_0
Level of Significance (α)	Probability of Type I error

If you want, I can also explain it in **very simple language** or in **Marathi** 😊

13) Discuss the importance of sampling in statistical analysis. Explain situations in business where Stratified Sampling is more appropriate than Random Sampling.

Here is a **clear and exam-ready explanation** of your question:

✅ Importance of Sampling in Statistical Analysis

Sampling is the process of selecting a small group (sample) from a large population to study and draw conclusions about the whole population.

Why sampling is important:

1. **Saves Time**
    - Studying the entire population (census) takes a lot of time.
    - Sampling gives faster results.
2. **Cost Effective**
    - Collecting data from everyone is very expensive.
    - Sampling reduces research and data collection costs.
3. **Practical for Large Populations**
    - When the population is too large, sampling is the only feasible method.
4. **Better Data Quality**
    - Researchers can focus carefully on a smaller group.
    - Reduces errors in data collection.
5. **Makes Statistical Analysis Possible**
    - Many statistical methods (like hypothesis testing, confidence intervals) require sample data.

Example:

A company wants to know customer satisfaction but has 1 million customers. Surveying everyone is difficult → so they survey 1,000 customers.

✅ Stratified Sampling vs Random Sampling

1. Simple Random Sampling

- Every individual has an **equal chance** of being selected.
- Selection is completely random.

Example:
Picking customer names randomly from a list.

2. Stratified Sampling

- Population is first divided into **subgroups (strata)** based on shared characteristics.
- Then random samples are taken **from each group**.

Common strata:

- Age groups
- Income levels
- Departments
- Regions
- Gender

✅ When Stratified Sampling is More Appropriate in Business

Stratified sampling is better when the population is **not uniform (heterogeneous)** and contains **clearly different groups**.

Important Business Situations:

1. Market Research by Age Group

A company launching a product wants opinions from:

- Teenagers
- Adults
- Senior citizens

✅ Stratified sampling ensures representation from **all age groups**.

---

2. Employee Satisfaction Survey

In a large company with:

- Managers
- Executives
- Workers

✅ Each group has different opinions, so stratified sampling gives balanced results.

---

3. Income-based Customer Analysis

A bank wants to study:

- Low-income customers
- Middle-income customers
- High-income customers

✅ Stratified sampling ensures that each income group is included.

---

4. Regional Sales Analysis

A company operating in:

- Urban areas
- Semi-urban areas
- Rural areas

✅ Stratified sampling ensures data from all regions.

✅ **Why Stratified Sampling is Better than Simple Random Sampling**

Point	Random Sampling	Stratified Sampling
Representation	May miss some groups	Ensures all groups are represented
Accuracy	Moderate	Higher accuracy
Usefulness	Good for uniform populations	Best for diverse populations

---

✅ **Final Short Conclusion (for exams)**

Stratified sampling is more suitable than random sampling in business situations where the population consists of different subgroups such as age, income, region, or job roles. It ensures fair representation, improves accuracy, and leads to better decision-making.

If you want, I can also give you **short 5-mark answer** or **Marathi version** 😊

14) A manufacturer claims that the average life of its bulbs is 1,000 hours. A sample of 40 bulbs shows a mean life of 960 hours with population standard deviation 80 hours. Test the claim at 5% level of significance. State: (a) Null and alternative hypothesis (b) Test statistic (c) Conclusion

Let's test the claim step by step.

Given:

- Claimed mean life = $\mu_0 = 1000$ hours
- Sample size = $n = 40$
- Sample mean = $\bar{x} = 960$ hours
- Population standard deviation = $\sigma = 80$ hours
- Level of significance = $\alpha = 0.05$

We use a **Z-test** because σ is known and n is large.

(a) Null and Alternative Hypothesis

Since the manufacturer claims the mean is 1000 hours, we test if the true mean is **different** from 1000.

$$H_0 : \mu = 1000 \quad \text{(average life is 1000 hours)}$$

$$H_1 : \mu \neq 1000 \quad \text{(average life is not 1000 hours)}$$

(This is a **two-tailed test**.)

(b) Test Statistic

Z-test formula:

$$
Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}
$$

Substitute values:

$$
Z = \frac{960 - 1000}{80 / \sqrt{40}}
= \frac{-40}{80 / \sqrt{40}}
$$

First compute denominator:

$$
80 / \sqrt{40} \approx 80 / 6.3249 \approx 12.65
$$

So:

$$
Z \approx \frac{-40}{12.65} \approx -3.16
$$

At ($\alpha = 0.05$) for a **two-tailed test**, critical Z values are:

$$
Z_{crit} = \pm 1.96
$$

(c) Conclusion

- Calculated ($Z = -3.16$)
- This lies **outside** the range (-1.96) to (+1.96).
- So, **reject (H_0)** at 5% level of significance.

✅ **Conclusion (in words):**
There is sufficient statistical evidence at the 5% level to reject the manufacturer's claim that the average life of the bulbs is 1,000 hours. The sample suggests that the true average life is **significantly different (and in fact lower)** than 1,000 hours.

15) From a large population with σ = 15, a sample of size 36 gives a mean of 120. Find the 95% confidence interval and comment whether a population mean of 125 is plausible

Given:

- Population standard deviation (\sigma = 15)
- Sample size (n = 36)
- Sample mean (\bar{x} = 120)
- Confidence level = **95%** → (Z = 1.96)

Step 1: Standard Error (SE)

[
SE = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5
]

Step 2: Confidence Interval Formula

[
\text{CI} = \bar{x} \pm Z \times SE
]

[
= 120 \pm (1.96 \times 2.5)
]
[

= 120 \pm 4.9
]

✅ **95% Confidence Interval**

[
(115.1,;124.9)
]

Comment on population mean of 125

- The value **125 is NOT inside** the confidence interval ((115.1,;124.9)).
- Therefore, **125 is not a plausible (likely) value** for the population mean at the 95% confidence level.

Final Answer:

- **95% Confidence Interval:** ((115.1,;124.9))
- **Conclusion:** A population mean of **125 is not plausible** at the 95% confidence level.

16) What do the regression coefficients represent in a simple linear regression model? Explain the interpretation of the slope and intercept with suitable examples.

Here's a **detailed, easy-to-understand explanation** of regression coefficients in **Simple Linear Regression**, with a **large practical example**.

---

1. Simple Linear Regression Model

The general equation is:

$$
Y = a + bX
$$

Where:

- (Y) = Dependent (response) variable
- (X) = Independent (predictor) variable
- (a) = **Intercept**
- (b) = **Slope (regression coefficient)**

---

2. What Do Regression Coefficients Represent?

Regression coefficients tell us how the dependent variable (Y) changes when the independent variable (X) changes.

There are two coefficients:

(a) Intercept (a)

- It represents the **value of Y when X = 0**.
- It is the starting point or baseline value.

(b) Slope (b)

- It represents the **change in Y for a one-unit change in X**.
- It shows the **direction** and **strength** of the relationship.

---

3. Interpretation of Slope and Intercept (with Detailed Examples)

✅ Example 1: Sales vs Advertising Budget

Suppose a company studies how sales depend on advertising.

Let the regression equation be:

$$
\text{Sales} = 50 + 2.5 \times \text{Advertising Spend}
$$

Where:

- Sales = in **thousands of units**
- Advertising Spend = in **thousands of rupees**

Intercept (a = 50)

Meaning:

If the company spends **0 rupees** on advertising, it will still sell **50,000 units**.

This represents **base sales** without advertising.

Slope (b = 2.5)

Meaning:

For every **₹1,000 increase** in advertising, sales increase by **2,500 units**.

So:

- If ad spend increases by 10 (₹10,000), sales increase by:
  [
  2.5 \times 10 = 25 \text{ thousand units}
  ]

✅ Example 2: Study Hours vs Exam Marks

A study finds the relationship:

[
\text{Marks} = 30 + 5 \times \text{Study Hours}
]

Intercept (a = 30)

Meaning:

If a student studies **0 hours**, expected marks are **30** (maybe from class knowledge).

Slope (b = 5)

Meaning:

For each additional **1 hour** of study, marks increase by **5 marks**.

Predictions:

Study Hours (X)	Predicted Marks (Y)
0	(30 + 5(0) = 30)
2	(30 + 5(2) = 40)
5	(30 + 5(5) = 55)
10	(30 + 5(10) = 80)

✅ Example 3: Temperature vs Ice Cream Sales (Large Example)

A shop wants to understand how temperature affects ice cream sales.

The equation is:

$$
\text{Sales} = 100 + 20 \times \text{Temperature}
$$

Where:

- Sales = Number of ice creams sold per day
- Temperature = in °C

Intercept (a = 100)

Meaning:

At **0°C**, the shop still sells **100 ice creams per day**.

This is **basic demand**.

Slope (b = 20)

Meaning:

For every **1°C increase in temperature**, sales increase by **20 ice creams per day**.

Predictions:

Temperature (°C)	Expected Sales
10	(100 + 20(10) = 300)
20	(100 + 20(20) = 500)
30	(100 + 20(30) = 700)
35	(100 + 20(35) = 800)

---

4. Direction of Relationship from Slope

Value of (b)	Meaning
(b > 0)	Positive relationship (Y increases as X increases)
(b < 0)	Negative relationship (Y decreases as X increases)
(b = 0)	No relationship

Example of Negative Slope:

$$
\text{Speed} = 80 - 2 \times \text{Traffic Density}
$$

Meaning:

As traffic increases by 1 unit, speed decreases by 2 km/hr.

---

5. Real-Life Business Example (Large Scenario)

A factory studies production cost vs number of units produced.

Regression equation:

$$
\text{Cost} = 5000 + 30 \times \text{Units Produced}
$$

Where:

- Cost in rupees
- Units Produced in units

Intercept = 5000

Meaning:

Even if the factory produces **zero units**, fixed costs like rent, electricity still cost ₹5,000.

Slope = 30

Meaning:

For every **1 extra unit produced**, the cost increases by **₹30** (variable cost per unit).

Prediction Table:

Units Produced	Predicted Cost
0	₹5000
50	₹5000 + 30(50) = ₹6500
100	₹5000 + 30(100) = ₹8000
200	₹5000 + 30(200) = ₹11,000

---

6. Final Summary (Exam-Ready)

- The **intercept** represents the expected value of the dependent variable when the independent variable is zero.
- The **slope** represents how much the dependent variable changes for a one-unit change in the independent variable.
- These coefficients help in understanding relationships and making predictions.

If you want, I can also give you a **short 5-mark or 10-mark answer** for exams 😊

17) Describe the Least Squares Method for estimating a regression line. Explain how the formulas for slope (b_1) and intercept (b_0) are derived.

Here is a **detailed, clear, and exam-ready explanation** of the **Least Squares Method** and the **derivation of formulas for slope and intercept**.

---

1. What is the Least Squares Method?

The **Least Squares Method** is a statistical technique used to find the **best-fitting straight line** (regression line) for a given set of data.

This method chooses the line such that the **sum of the squares of the vertical distances (errors)** between the actual values and the predicted values is **minimum**.

In simple words:

It finds the line that best represents the relationship between X and Y by minimizing errors.

---

2. Regression Line Equation

The equation of a simple linear regression line is:

[
Y = b_0 + b_1X
]

Where:

- (Y) = Dependent variable
- (X) = Independent variable
- (b_0) = Intercept (constant)
- (b_1) = Slope (regression coefficient)

---

3. Basic Idea of Least Squares

Suppose we have observations:

$$
(x_1,y_1), (x_2,y_2), \ldots, (x_n,y_n)
$$

The **predicted value** of (Y) is:

$$
\hat{Y} = b_0 + b_1 X
$$

Error (or residual) for each observation:

$$
e_i = y_i - \hat{Y_i} = y_i - (b_0 + b_1 x_i)
$$

The **sum of squared errors**:

$$
S = \sum (y_i - (b_0 + b_1 x_i))^2
$$

The goal of least squares is to **minimize S**.

---

4. Derivation of Formulas for (b_0) and (b_1)

We minimize (S) using partial derivatives.

Step 1: Take partial derivatives and set them equal to zero

Differentiate with respect to (b_0):

$$
\frac{\partial S}{\partial b_0} = -2 \sum (y_i - b_0 - b_1 x_i) = 0
$$

Differentiate with respect to (b_1):

$$
\frac{\partial S}{\partial b_1} = -2 \sum x_i (y_i - b_0 - b_1 x_i) = 0
$$

Divide by -2:

$$
\sum (y_i - b_0 - b_1x_i) = 0 \quad ...(1)
$$

$$
\sum x_i (y_i - b_0 - b_1x_i) = 0 \quad ...(2)
$$

Step 2: Expand the equations

From (1):

$$
\sum y_i = nb_0 + b_1 \sum x_i \quad ...(A)
$$

From (2):

$$
\sum x_i y_i = b_0 \sum x_i + b_1 \sum x_i^2 \quad ...(B)
$$

These are called the **normal equations**.

Step 3: Solve for (b_1) (Slope)

From (A) and (B), the slope formula is derived:

$$
b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}
$$

Step 4: Solve for (b_0) (Intercept)

Once (b_1) is found:

$$
b_0 = \bar{y} - b_1 \bar{x}
$$

Where:

$$
\bar{x} = \frac{\sum x}{n}, \quad \bar{y} = \frac{\sum y}{n}
$$

---

5. Final Derived Formulas

Slope:

$$
b_1 = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}
$$

Intercept:

$$
b_0 = \bar{y} - b_1 \bar{x}
$$

---

6. Small Numerical Example

Given data:

X	Y
1	2
2	4
3	5

Calculate:

$$
n = 3
$$

$$
\sum x = 6,; \sum y = 11,; \sum xy = 25,; \sum x^2 = 14
$$

Slope:

$$
b_1 = \frac{3(25) - (6)(11)}{3(14) - 6^2}
= \frac{75 - 66}{42 - 36}
= \frac{9}{6} = 1.5
$$

Intercept:

$$
\bar{x} = 2,; \bar{y}= \frac{11}{3}
$$

$$
b_0 = \frac{11}{3} - (1.5)(2) = 3.67 - 3 = 0.67
$$

So regression equation:

$$
Y = 0.67 + 1.5X
$$

---

7. Short Exam Conclusion

The Least Squares Method finds the best-fitting regression line by minimizing the sum of squared errors. The formulas for slope and intercept are derived by differentiating the error sum and solving the resulting normal equations.

If you want, I can provide a **short 5-mark version** or **Marathi explanation** too 😊

18) For a regression model, the following values are given: Total Sum of Squares (TSS) = 400 Explained Sum of Squares (ESS) = 280 Calculate: (a) R-squared (b) Interpret the result in context

Given:

- **Total Sum of Squares (TSS) = 400**
- **Explained Sum of Squares (ESS) = 280**

(a) Calculate (R^2) (Coefficient of Determination)

Formula:

$$R^2 = \frac{ESS}{TSS}$$

$$R^2 = \frac{280}{400} = 0.70$$

✅ **Answer:**

$$R^2 = 0.70 \text{ or } 70\%$$

(b) Interpretation of the Result

An (R^2) value of **0.70** means:

70% of the total variation in the dependent variable is explained by the regression model.

In simple words:

The independent variable(s) used in the model explain **most of the changes** in the dependent variable, so the model has a **good fit**.

Example-style interpretation:

If this model predicts **sales** based on **advertising**, then:

- 70% of the variation in sales is explained by advertising.
- The remaining 30% is due to other factors not included in the model.

Final Answers:

(a) ($R^2 = 0.70$)
(b) The model explains **70% of the variation**, indicating a **strong relationship and good explanatory power**.

19) The following data relate advertising expenditure (X in ₹'000) and sales (Y in ₹'000): X 2 3 5 7 9 Y 4 5 7 10 15 Calculate: (a) The regression line of Y on X using the least squares method (b) Predict sales when advertising expenditure is ₹6,000

Let's do this step by step using **least squares**.

We're given:

X (₹'000) 2 3 5 7 9

Y (₹'000) 4 5 7 10 15

We want the regression line of **Y on X**:

$$\hat{Y} = b_0 + b_1 X$$

(a) Find the regression line of Y on X

Step 1: Compute needed sums

Let's make a working table:

X	Y	X^2	X·Y
2	4	4	8
3	5	9	15
5	7	25	35
7	10	49	70
9	15	81	135

Now sum each column:

- $(n = 5)$
- $(\sum X = 2+3+5+7+9 = 26)$
- $(\sum Y = 4+5+7+10+15 = 41)$
- $(\sum X^2 = 4+9+25+49+81 = 168)$
- $(\sum XY = 8+15+35+70+135 = 263)$

Step 2: Formula for slope (b_1)

$$
b_1 = \frac{n\sum XY - (\sum X)(\sum Y)}{n\sum X^2 - (\sum X)^2}
$$

Substitute:

$$
b_1 = \frac{5\cdot263 - 26\cdot41}{5\cdot168 - 26^2}
= \frac{1315 - 1066}{840 - 676}
= \frac{249}{164} \approx 1.518
$$

Step 3: Formula for intercept (b_0)

$$
\bar{X} = \frac{\sum X}{n} = \frac{26}{5} = 5.2,\quad
\bar{Y} = \frac{\sum Y}{n} = \frac{41}{5} = 8.2
$$

$$
b_0 = \bar{Y} - b_1 \bar{X}
= 8.2 - 1.518\times 5.2 \approx 0.305
$$

(Exactly, $b_0 = \frac{25}{82} \approx 0.305$)

✅ **Regression line of Y on X:**

$$
\boxed{\hat{Y} = 0.305 + 1.518X} \quad \text{(approximately)}
$$

(b) Predict sales when advertising = ₹6,000

Advertising is in **₹'000**, so:

- ₹6,000 ⇒ $(X = 6)$

Use the regression equation:

$$
\hat{Y} = 0.305 + 1.518 \times 6
\approx 0.305 + 9.108
\approx 9.415
$$

So predicted sales:

- $\hat{Y} \approx 9.415$ (in ₹'000)

👉 **Predicted sales ≈ ₹9,400–₹9,500 (about ₹9,415).**

Final Answers

(a) Regression line:

$$
\hat{Y} \approx 0.305 + 1.518X
$$

(b) For advertising ₹6,000 ⇒ **Predicted sales ≈ ₹9,415**.

20) . Discuss the Moving Averages method of forecasting. Explain its advantages, limitations, and suitable business applications.

Here's a clear, detailed, and exam-ready explanation of the **Moving Averages Method of Forecasting**:

---

1. What is the Moving Averages Method?

The **Moving Averages method** is a **time series forecasting technique** used to smooth out short-term fluctuations and highlight long-term trends.

It calculates the average of a fixed number of recent observations and uses this average as the forecast for the next period.

In simple words:
It predicts future values by taking the average of the most recent data points.

---

2. How Moving Averages Work (Concept)

Suppose we use a **3-period moving average**.

Example data (monthly sales):

Month	Sales
Jan	100
Feb	110
Mar	120
Apr	130
May	140

To forecast **June** using a 3-month moving average:

[
\text{Forecast for June} = \frac{Sales_{Mar} + Sales_{Apr} + Sales_{May}}{3}

$$= \frac{120 + 130 + 140}{3} = 130$$

---

3. Steps in Moving Averages Method

1. Choose the number of periods (e.g., 3, 4, or 5).
2. Add the values of the selected periods.
3. Divide by the number of periods to find the average.
4. Use this average as the forecast for the next period.
5. As new data becomes available, drop the oldest value and add the latest value.

This "moving" of periods gives the method its name.

---

4. Types of Moving Averages

(a) Simple Moving Average (SMA)

All observations are given **equal weight**.

Formula:

$$SMA = \frac{Y_1 + Y_2 + ... + Y_n}{n}$$

(b) Weighted Moving Average (WMA)

Different weights are given to recent data.

Example weights: 3, 2, 1

---

5. Advantages of Moving Averages

1. **Simple to Understand and Use**
   Easy calculation, no complex formulas.
2. **Reduces Random Fluctuations**
   Smooths irregular movements in sales or demand.

3. **Useful for Short-Term Forecasting**
   Effective when the market is stable.
4. **Less Impact of Sudden Changes**
   Minimizes the impact of abnormal spikes or dips.

---

6. Limitations of Moving Averages

1. **Ignores Trend and Seasonality**
   Cannot handle strong seasonal patterns.
2. **Lag Effect**
   Forecasts lag behind actual changes because only past data is used.
3. **Choice of Period is Subjective**
   Wrong period selection can reduce accuracy.
4. **Not Suitable for Rapidly Changing Markets**
   Performs poorly when demand changes suddenly.

---

7. Suitable Business Applications

1. Sales Forecasting

Used by retailers to predict next month's sales.

2. Inventory Management

Helps companies decide how much stock to keep.

3. Financial Market Analysis

Used in stock price trend analysis.

4. Production Planning

Helps factories plan production level.

5. Demand Forecasting

Used in supermarkets, FMCG companies, and e-commerce.

---

8. Real Business Example

A supermarket wants to forecast the demand for milk.

Past daily demand (liters):
100, 110, 120, 115, 125

3-day moving average for next day:

$$
\frac{120 + 115 + 125}{3} = 120
$$

So, the store plans for **120 liters**.

---

9. Short Exam-Style Conclusion

The Moving Averages method is a simple and effective forecasting technique that uses the average of recent data to predict future values. It is most suitable for businesses with stable demand patterns but is limited in handling trends and seasonal variations.

If you want, I can also provide a **short 5-mark version** of this answer 😊

21) Two machines produce items with the following statistics: Machine Mean output Standard deviation A 50 6 B 48 4 Calculate the Coefficient of Variation (CV) for both machines and state which machine is more consistent.

To solve this, we use the **Coefficient of Variation (CV)** formula:

$$
CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100
$$

✅ **Given Data**

Machine	Mean (μ)	Standard Deviation (σ)
A	50	6
B	48	4

(a) CV for Machine A

$$
CV_A = \frac{6}{50} \times 100 = 12\%
$$

(b) CV for Machine B

$$
CV_B = \frac{4}{48} \times 100 = 8.33\% \;(\text{approx})
$$

(c) Which machine is more consistent?

- A **lower CV** means **more consistency**.
- Machine **B** has a lower CV (8.33%) than Machine A (12%).

✅ **Conclusion:**
Machine B is more consistent because it has less variation relative to its mean.

Final Answers:

Machine	CV
A	**12%**
B	**8.33%**

More consistent machine: Machine B ✅

22) A company recorded the number of defective bulbs produced by two machines in a week: Day Machine A Machine B Mon 6 4 Tue 9 5 Wed 5 3 Thu 9 9 Fri 3 6 Using your understanding of measures of dispersion, determine which machine shows greater consistency in quality.

Let's analyze **consistency using measures of dispersion** (Standard Deviation and Coefficient of Variation).

Given Data

Day	Machine A	Machine B
Mon	6	4
Tue	9	5
Wed	5	3
Thu	9	9
Fri	3	6

Step 1: Calculate Mean

Machine A:

$$
\text{Mean}_A = \frac{6+9+5+9+3}{5} = \frac{32}{5} = 6.4
$$

Machine B:

$$
\text{Mean}_B = \frac{4+5+3+9+6}{5} = \frac{27}{5} = 5.4
$$

Step 2: Calculate Standard Deviation

Machine A

X	X − Mean (6.4)	(X − Mean)²
6	-0.4	0.16
9	2.6	6.76
5	-1.4	1.96
9	2.6	6.76
3	-3.4	11.56

$$
\sum (X-\bar X)^2 = 0.16+6.76+1.96+6.76+11.56 = 27.2
$$

Variance:

$$
\sigma_A^2 = \frac{27.2}{5} = 5.44
$$

Standard Deviation:

$$
\sigma_A = \sqrt{5.44} \approx 2.33
$$

Machine B

X	X − Mean (5.4)	(X − Mean)²
4	-1.4	1.96
5	-0.4	0.16
3	-2.4	5.76
9	3.6	12.96
6	0.6	0.36

$$
\sum (X-\bar X)^2 = 1.96+0.16+5.76+12.96+0.36 = 21.2
$$

Variance:

$$
\sigma_B^2 = \frac{21.2}{5} = 4.24
$$

Standard Deviation:

$$
\sigma_B = \sqrt{4.24} \approx 2.06
$$

Step 3: Coefficient of Variation (CV)

$$
CV = \frac{\sigma}{\text{Mean}} \times 100
$$

Machine A:

$$
CV_A = \frac{2.33}{6.4} \times 100 \approx 36.41\%
$$

Machine B:

$$
CV_B = \frac{2.06}{5.4} \times 100 \approx 38.15\%
$$

✅ **Final Comparison**

Machine	Mean	Standard Deviation	CV (%)
A	6.4	2.33	**36.41%**
B	5.4	2.06	38.15%

✅ **Conclusion**

Since **Machine A has a lower Coefficient of Variation (36.41%)** compared to Machine B (38.15%), it shows **greater consistency in quality**.

✅ **Final Answer:**

Machine A is more consistent in quality.

If you want a shorter exam-style answer, just tell me 😊

23) A company wants to know whether advertising expenditure (X) affects sales (Y). The data is: X (₹'000) 5 10 15 20 25 Y (Units) 50 55 65 70 75 Calculate the correlation coefficient and interpret whether there is a strong relationship between advertising and sales.

Let's calculate the **correlation coefficient (r)** step by step.

Given Data

X (₹'000)	**5**	**10**	**15**	**20**	**25**
Y (Units)	50	55	65	70	75

Step 1: Prepare the working table

We will use the **Karl Pearson correlation formula**.

X	**Y**	**X^2**	**Y^2**	**XY**
5	50	25	2500	250

X	Y	X^2	Y^2	XY
10	55	100	3025	550
15	65	225	4225	975
20	70	400	4900	1400
25	75	625	5625	1875

Now calculate the totals:

- $\sum X = 5+10+15+20+25 = 75$
- $\sum Y = 50+55+65+70+75 = 315$
- $\sum X^2 = 25+100+225+400+625 = 1375$
- $\sum Y^2 = 2500+3025+4225+4900+5625 = 20275$
- $\sum XY = 250+550+975+1400+1875 = 5050$
- $n = 5$

Step 2: Use the Karl Pearson correlation formula

$$
r = \frac{n\sum XY - (\sum X)(\sum Y)}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}
$$

Substitute values:

Numerator:

$$
5(5050) - (75)(315) = 25250 - 23625 = 1625
$$

Denominator:

$$
\sqrt{[5(1375) - 75^2][5(20275) - 315^2]}
$$

First bracket:

$$
5(1375) - 75^2 = 6875 - 5625 = 1250
$$

Second bracket:

$$
5(20275) - 315^2 = 101375 - 99225 = 2150
$$

Now:

$$
\sqrt{1250 \times 2150} = \sqrt{2687500} \approx 1639.05
$$

Final value of r:

$$
r = \frac{1625}{1639.05} \approx 0.99
$$

✅ Final Answers

(a) Correlation Coefficient:

$$
\boxed{r \approx 0.99}
$$

(b) Interpretation

- The value of **r is very close to +1**.
- This indicates a **very strong positive correlation** between advertising expenditure and sales.

In simple words:

As advertising spending increases, sales also increase strongly.

✅ **Conclusion:** There is a **strong positive relationship** between advertising and sales.

If you want, I can also show you a **shortcut method** for this type of question 😊

24) Explain the components of a Time Series (Trend, Seasonal, Cyclical, Irregular). Give business examples for each component. Production 10–20 20–30 30–40 40–50 50–60 Frequency 5 8 12 9 6 Calculate: (a) Mean production (b) Standard deviation

Let's break it into **two parts**:
1⃣ Theory – components of a time series
2⃣ Numerical – mean and standard deviation of production

---

1) Components of a Time Series (with business examples)

A **Time Series** is data recorded over time (daily, monthly, yearly, etc.).
Example: monthly sales, daily stock prices, yearly production, etc.

A time series usually has **four components**:

1. Trend (T)

- **Meaning:** Long-term **upward or downward movement** in the data over many years.
- It shows the **general direction** of the series.
- It is **smooth**, not worried about short-term ups and downs.

Business examples:

- A company's sales increasing steadily over 10 years.

- Long-term increase in production capacity in a factory.
- Gradual rise in number of internet users over years.

---

2. Seasonal Variation (S)

- **Meaning:** Regular, **repeating pattern within a year**, quarter, month, week, or even within a day.
- Caused by **climate, festivals, customs, habits**, etc.

Business examples:

- Ice-cream sales higher in **summer**, lower in winter.
- Garment sales increase during **Diwali / Eid / Christmas**.
- More passengers in buses/trains during **office hours** (morning & evening).

---

3. Cyclical Variation (C)

- **Meaning:** Long-term **up-and-down movements** around the trend line, usually lasting **more than one year**.
- Related to **business cycles**:
  Boom → Prosperity → Recession → Depression → Recovery

Business examples:

- Automobile or real estate sales rise during **economic boom**, fall during **recession**.
- Company profits going up and down with **overall economy** over 5–10 years.

---

4. Irregular / Random Variation (I)

- **Meaning:** Unpredictable, **random fluctuations** caused by unusual or accidental factors.
- They **do not repeat regularly** and cannot be forecast accurately.

Business examples:

- Sudden drop in production due to **strike**, **flood**, **fire**, **machinery breakdown**.
- Sharp increase in sales due to **one-time advertisement campaign** or **celebrity endorsement**.
- COVID-19 impact on travel, cinema, hotels.

So, a time series can be thought of as:

Observed value = Trend + Seasonal + Cyclical + Irregular components

---

2) Mean and Standard Deviation of Production

You're given grouped production data:

Production (Class)	Frequency
10–20	5
20–30	8
30–40	12
40–50	9
50–60	6

We'll treat the class marks (mid-points) as representative values.

Step 1: Find midpoints and f×x

Class	Midpoint (x)	f	f·x
10–20	15	5	75
20–30	25	8	200
30–40	35	12	420
40–50	45	9	405

Class	Midpoint (x)	f	f·x
50–60	55	6	330

Now sum:

- Total frequency:
  (N = 5 + 8 + 12 + 9 + 6 = 40)
- Sum of f·x:
  (\sum fx = 75 + 200 + 420 + 405 + 330 = 1430)

(a) Mean Production

[
\bar{x} = \frac{\sum fx}{N} = \frac{1430}{40} = 35.75
]

✅ **Mean production ≈ 35.75 units** (you can write 35.8 if rounding to one decimal).

(b) Standard Deviation

We use:

[
\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}
]

First find ((x - \bar{x})^2) and multiply by f:

Mean (\bar{x} = 35.75)

x	f	x − 35.75	(x − 35.75)²	f·(x − 35.75)²
15	5	-20.75	430.5625	2152.8125
25	8	-10.75	115.5625	924.5
35	12	-0.75	0.5625	6.75

x	f	x − 35.75	(x − 35.75)²	f·(x − 35.75)²
45	9	9.25	85.5625	770.0625
55	6	19.25	370.5625	2223.375

Now sum:

$$\sum f(x - \bar{x})^2 = 2152.8125 + 924.5 + 6.75 + 770.0625 + 2223.375 = 6077.5$$

Now variance:

$$\sigma^2 = \frac{6077.5}{40} = 151.9375$$

Standard deviation:

$$\sigma = \sqrt{151.9375} \approx 12.33$$

✅ **Standard deviation ≈ 12.33 units**

Final Answers

Time Series Components:

- **Trend:** Long-term direction (e.g., long-term growth in sales).
- **Seasonal:** Regular within-year pattern (e.g., festival/season-based sales).
- **Cyclical:** Multi-year business cycle movements.
- **Irregular:** Random shocks (e.g., strikes, disasters).

Numerical:

- (a) **Mean production ≈ 35.75 units**
- (b) **Standard deviation ≈ 12.33 units**

If you want, I can show a **shortcut (assumed mean) method** for SD also, like you'd write in the exam.